

SPENCER BIALEK, PH.D.

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EDUCATION

University of Victoria

Ph.D. Physics & Astronomy

Victoria, BC

Jan. 2020 – Dec. 2023

- Relevant research work: Applied Deep Learning, Cloud Computing, Python Development, Big Data

University of Victoria

M.Sc. Physics & Astronomy

Victoria, BC

Sep. 2017 – Dec. 2019

- Relevant research work: Python development, Applied Machine Learning
- Relevant coursework: Machine Learning, Neural Networks, Distributed Computing

University of Victoria

B.Sc. Honours Physics & Astronomy with Co-op Distinction

Victoria, BC

Sep. 2011 – May 2017

SKILLS AND QUALIFICATIONS

Programming Languages: Python (11 years), C and C++ (2+ years), Java (1 year), Octave (1 year), HTML/CSS (1 year), JavaScript/Rust (4+ months), MATLAB (2+ months)

Software & Tools: Git, GitHub, Docker, LLM-assisted coding (Claude Code, Codex), OpenStack, Slurm, PyCharm, Weights & Biases, VS Code, LaTeX, Excel

Python Libraries & Applications: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, PyTorch, TensorFlow, Transformers, Astropy, Jupyter, Label Studio, Streamlit, Dash

Methods: Deep learning, self-supervised learning, anomaly detection, acoustic-data analysis, natural language processing, large language models, retrieval-augmented generation

Communication: Science writing, outreach, public speaking, data visualization, tutorials and technical documentation

HONOURS, SCHOLARSHIPS, AND AWARDS

Anthropic AI For Science Award

2026

President's Research Scholarship

2020 – 2023

NSERC Postgraduate Scholarship – Doctoral

2020 – 2023

NSERC CREATE Scholarship

2020 – 2022

Boehm-Hesser Graduate Research Excellence in Astronomy Award

2020

UVic Graduate Award

2017-2020

UVic Fellowship

2017

Boehm Family Award for Excellence in Science

2016

Robert S. Evans Memorial Scholarship

2016

2nd place in poster fair at Canadian Undergraduate Physics Conference

2016

Outstanding Co-op Work Term Report

2014

WORK/INDUSTRY EXPERIENCE

Machine Learning Specialist

Victoria, BC

Ocean Networks Canada

July 2024 – Present

- **Whale autodetection:** Advanced and evaluated automated whale-call detection workflows for hydrophone data, supporting scalable screening of acoustic recordings and expert review of candidate detections
- **Human-in-the-loop ML:** Created a custom spectrogram labeling and verification app for domain experts to label data, review machine learning predictions, and support model evaluation
- **Hydrophone anomaly detection:** Developed automated self-supervised pipelines for detecting anomalies in hydrophone data, improving consistency and efficiency of acoustic-data monitoring workflows
- **Scientific literature analysis:** Designed and implemented an LLM/RAG-based tool for analyzing large scientific literature corpora and generating structured reports from user-defined topics and queries
- **User-facing software:** Built scientist-facing applications, tutorials, and documentation to make custom machine learning tools accessible to non-ML users

Research Internship

Canada-France-Hawaii Telescope

Waimea, HI

May 2022 – October 2022

- Helped build simulation software using PyTorch to generate training and test sets of observations of stars with atmospheric turbulence and noise
- Developed a novel machine learning system which infers in real-time the turbulence- and noise-free image from a video sequence of short-exposure images over a wide field
- Developed into the main Ph.D. project and first-author DanceCam paper in Monthly Notices of the Royal Astronomical Society (2024); project website: dancecam.info

Research and Software Development Internship

Limbic Media

Victoria, BC

May 2018 – Sep. 2018

- Curated and reduced a training set of audio spectrogram data from thousands of songs
- Implemented a neural network-based analysis of music boundary detection for the Aurora sound-responsive lighting system, doubling the performance over competing methods

Research Assistant Co-op

NRC Herzberg Astronomy and Astrophysics

Victoria, BC

May 2017 – Sep. 2017

- Created a machine learning pipeline capable of deriving fundamental properties of stars including temperature, surface gravity, and metallicity

RESEARCH EXPERIENCE

University of Victoria, Ph.D. Physics & Astronomy

Victoria, BC

Project: StarUnlink

2021 – 2022

- Pioneered a deep learning framework using CNN and U-Net architectures to detect when a stellar spectrum is contaminated by an orbiting communications satellite from companies like Starlink, and either accurately predict the star's properties despite the contamination, or remove the contamination from the data for downstream tasks
- Joined the International Astronomical Union's Centre for the Protection of the Dark and Quiet Sky from Satellite Constellation Interference (IAU-CPS)
- Primary authorship of peer-reviewed publication and presentation at IAU Symposium
- Secured \$90,000 in research grants during doctoral research

Project: Automatic Galaxy Morphology Classification

2021 – 2023

- Created a contrastive self-supervised method for automating the classification of galaxy morphologies from the UNIONS survey using learned neural network features. The "galaxy recommender" system finds the most similar galaxies to a galaxy of interest, helping to discover new examples of rare types of galaxies
- Supervised a research team developing and applying neural network methodologies for rare galaxy discovery

University of Victoria, M.Sc. Physics & Astronomy

Victoria, BC

Project: StarNet

2017 – 2019

- Built an end-to-end cloud-computing pipeline that generates a large set of (several hundred thousand) synthetic stellar spectra, trains a deep ensemble machine learning model on them, and predicts stellar parameters and chemical abundances – along with uncertainties – for several spectroscopic surveys in seconds
- Evaluated the quality of several commonly used libraries of synthetic stellar spectra to provide recommendations to the astronomical community
- Primary authorship of peer-reviewed publication and poster at national conference

TEACHING EXPERIENCE

Teaching Assistant

Victoria, BC

University of Victoria

2017 – 2024

- Taught the laboratory sections for several courses at the University of Victoria: ASTR 101, ASTR 102, ASTR 150, ASTR 201, and PHYS 216
- Lectured, demonstrated laboratory procedures, and facilitated discussions
- Helped organize and prepare the teaching materials for other teaching assistants

PEER-REVIEWED PUBLICATIONS

1. **Bialek, S.**, Bertin, E., Fabbro, S., Bouy, H., Rivet, J.-P., Lai, O., & Cuillandre, J.-C. (2024). "DanceCam: Atmospheric turbulence mitigation in wide-field astronomical images with short-exposure video streams." *Monthly Notices of the Royal Astronomical Society*, 531(1), 403–421. doi:10.1093/mnras/stae1018.
2. **Bialek, S.**, Lucatello, S., Fabbro, S., Yi, K. M., & Venn, K. A. (2023). "StarUnLink: Identifying and mitigating signals from communications satellites in stellar spectral surveys." *Monthly Notices of the Royal Astronomical Society*, 524(1), 529–541.
3. O'Briain, T., Ting, Y. S., Fabbro, S., Yi, K. M., Venn, K., & **Bialek, S.** (2021). "Cycle-StarNet: Bridging the Gap between Theory and Data by Leveraging Large Data Sets." *The Astrophysical Journal*, 906(2), 130.
4. **Bialek, S.**, Fabbro, S., Venn, K. A., Kumar, N., O'Briain, T., & Yi, K. M. (2020). "Assessing the performance of LTE and NLTE synthetic stellar spectra in a machine learning framework." *Monthly Notices of the Royal Astronomical Society*, 498(3), 3817–3834.
5. Fabbro, S., Venn, K. A., O'Briain, T., **Bialek, S.**, Kielty, C. L., Jahandar, F., & Monty, S. (2018). "An application of deep learning in the analysis of stellar spectra." *Monthly Notices of the Royal Astronomical Society*, 475(3), 2978–2993.

POSTER AND PODIUM PRESENTATIONS

1. **Bialek, S.** "Automated Detection of Anomalous Hydrophone Data Using Self-Supervised Machine Learning." Oral presentation, International Conference on Underwater Acoustics (ICUA); Glasgow, UK (2026).
2. **Bialek, S.** Poster presentation, UACE 2025; Thessaloniki, Greece (2025).
3. **Bialek, S.**, Bertin, E., Fabbro, S. "Wide-field atmospheric turbulence mitigation using short-exposure video sequences and machine learning" Canadian Astronomical Instrumentation Workshop; Montreal, QC (2023).
4. O'Briain, T., Ting, Y. S., Fabbro, S., Yi, K. M., Venn, K., & **Bialek, S.** "Decreasing the Gap between Synthetic and Real Data... finally automated" Invited seminar at Columbia University; New York, NY (2021).
5. **Bialek, S.**, Fabbro, S., Venn, K. A., Kumar, N., O'Briain, T., & Yi, K. M. "Machine learning for stellar spectroscopy" Statistical Challenges in Modern Astronomy (SCMA) VII (2021).
6. **Bialek, S.**, Fabbro, S., Venn, K. A., Kumar, N., O'Briain, T., & Yi, K. M. "Machine learning for stellar spectroscopy" Summer School in Statistics for Astronomers; Penn State University (2021).
7. **Bialek, S.**, Fabbro, S., Venn, K. A. "StarNet: Harnessing the power of deep learning for stellar spectroscopy" European Astronomical Society (EAS) Annual Meeting (2020, 2021).
8. **Bialek, S.**, Fabbro, S., Venn, K. A. "Various StarNet related posters" Canadian Astronomical Society (CASCA) Annual Meeting (2017, 2018, 2019, 2020, 2021).
9. **Bialek, S.**, Fabbro, S., Venn, K. A. New Technologies for Canadian Observatories (NTCO) Annual Meeting. Invited talk (2019).

VOLUNTEERING AND PUBLIC OUTREACH

Nerd Nite Victoria	Victoria, BC
<i>Co-facilitator and co-host</i>	2019 – 2021
• Organized, co-hosted, and created promotional materials for the monthly science outreach event Nerd Nite.	
AstroCoffee	Victoria, BC
<i>Facilitator</i>	2019-2021
• Organized and facilitated weekly informal meetings for astronomers at UVic to discuss new research	
Black Rock Observatory	Black Rock Desert, NV
<i>Volunteer</i>	2017, 2019
• Operated several telescopes at Burning Man while teaching people about the celestial objects they were observing	
Bass Coast Festival	Merritt, BC
<i>Astronomer on Duty</i>	2017, 2018, 2019
• Facilitated discussions of astronomical phenomena while operating a telescope	